

Loss of chromosome 8p, and of *STC1*, distinguishes indolent from metastatic NSCLC: a negative-selection analysis of TCGA

Working draft — TCGA LUAD + LUSC (PanCancer Atlas). All analyses reproducible from scripts/ in this repository.

Abstract

Bulk primary tumours rarely carry a clean transcriptional signature of metastasis, because "metastatic" tumours differ from non-metastatic ones mainly in proliferation — a proxy for stage, not for metastatic capability. We reframed the question as **negative selection**: rather than asking what metastatic tumours *gain*, we asked what confirmed-indolent tumours *lose*. Using a stringent "true stage I" reference (stage I, N0/M0, recurrence-free ≥ 2 years) against tumours that progressed, we find that a chromosome **8p** deletion is present in ~9% of indolent tumours but almost absent (0.8%) in metastatic ones (subtype-adjusted CMH $q \approx 0.027$). Because 8p is lost as a whole arm, we used expression — controlling for both gene dosage and proliferation — to resolve the locus to **STC1** (stanniocalcin-1, 8p21.2), a secreted hypoxia-induced promoter of EMT and angiogenesis. High STC1 predicts significantly worse overall survival (median 44.0 vs 61.6 months; log-rank $p = 0.003$), an association independent of proliferation. We propose that retention of 8p/STC1 marks metastatic capability in NSCLC — an association-level hypothesis for validation in metastasis-enriched cohorts.

Background and hypothesis

Identifying genes that specifically mark metastasis (as opposed to general tumour aggression) is confounded in bulk primary tissue: metastatic primaries are simply more proliferative, and proliferation dominates every differential test.

We therefore inverted the logic. A tumour that presents at **stage I and remains recurrence-free for years** has demonstrably never acquired the ability to spread. If metastasis requires some cellular machinery, such indolent tumours may have **lost** it — for example by deep-deleting a gene they never needed.

Hypothesis. Genes required for (or permissive of) metastasis are **deletable in indolent tumours but retained in tumours that metastasize**. Equivalently: copy-number loss of metastasis machinery is a marker of indolent disease.

Methods

Cohort and data. TCGA lung adenocarcinoma (LUAD) and squamous-cell carcinoma (LUSC), PanCancer Atlas. mRNA expression (RSEM), discrete GISTIC copy number, and clinical data were obtained from cBioPortal (mirna_tcga.cbioportal). Distant- metastasis annotation was enriched with the TCGA **BCR Biotab** clinical supplements (mirna_tcga.biotab), unioning pathologic M1 with clinical M1 and follow-up new_tumor_event_type = 'Distant Metastasis'; this raised the confirmed distant-metastatic NSCLC set from ~33 to 126.

Groups. A **"true stage I"** indolent reference ($n = 218$) was defined as stage I at diagnosis, N0/M0, ≥ 2 -year follow-up with no distant or locoregional recurrence (biotab.true_stage_i_vs_distant). Progression-capable groups were distant metastasis ($n = 124$) and node-positive disease (N+, $n = 343$).

Negative-selection (deletion) screen. For each gene, deep-deletion frequency was compared between progressed and true-stage-I tumours with a **Cochran–Mantel–Haenszel test stratified by subtype** (associate.cmh_depletion_screen); genes depleted for deletion in progressed tumours are, equivalently,

deleted more in indolent tumours. BH-FDR across the deletable-gene set (script 23_true_stage_i_lost_genes.py).

Resolving the locus to a gene. Because 8p deletes as one arm, deletion cannot distinguish co-deleted genes. We tested each co-deleted gene's **expression** (progressed vs indolent, subtype-stratified van Elteren rank-sum, `associate.ranksum_screen`) against two confounders: (i) **dosage** — the same test restricted to tumours where the gene is *not* deleted; (ii) **proliferation** — strata of subtype × proliferation-tertile (`panels.PROLIFERATION_PANEL`) (script 24_chr8p23_which_gene.py).

Survival. Overall survival in NSCLC (~980 patients, 384 deaths). Association was tested with a vectorized **Cox score test** (`screen.cox_score_screen`), subtype-stratified and then subtype × proliferation-tertile stratified, with MKI67 as a proliferation positive control (script 25_stc1_survival_check.py). Kaplan–Meier curves and the two-group log-rank test (implemented directly) used a **within-subtype median split** of STC1 expression (script 26_stc1_km_curve.py).

Results

1. No positive metastasis signature in the bulk primary

Genome-wide differential expression and copy-number screens (scripts 17–22) found **no proliferation-independent signature of metastasis**. The entire nodal-metastasis expression signal was proliferation ($\approx 90\%$ of hits vanished after proliferation adjustment), and canonical EMT/invasion effectors (VIM, SNAI1/2, ZEB1/2, TWIST1, FN1, MMPs, LOXL2, ...) were not differentially expressed between node-positive and node-negative primaries (0/28 significant). Distant metastasis showed no signature in any layer.

2. Chromosome 8p is lost in indolent tumours and retained in metastatic ones

Against the true-stage-I reference, a co-deleted **chromosome 8p block** was the top — and only FDR-significant — signal:

Contrast	8p deleted in true stage I	in progressed	CMH OR	q
distant-met vs true-stage-I (124 vs 208)	9.1 %	0.8 % (1/124)	0.07	0.027
N+ vs true-stage-I (343 vs 208)	9.1 %	3.5 %	0.36	0.085

The signal was concordant across subtypes (LUAD distant-met $q \approx 0.09$; LUSC nodal $q \approx 0.20$) and reached significance on pooling. Indolent tumours delete 8p roughly one time in eleven; tumours that metastasized almost never do.

3. Expression resolves the arm to STC1

8p is lost as a whole arm (~190 co-deleted genes with identical q), so we used expression to identify the functional member. Most 8p genes are *lower* in progressed tumours (8p is tumour-suppressor-rich), but the retention-tracks- metastasis candidates are 8p genes **higher** in progressed that survive both confounders:

gene	up in progressed	dosage-independent	proliferation-adjusted	verdict
STC1	$z=+3.31$ ($p=0.001$)	$z=+2.12$ ($p=0.034$)	$z=+2.27$ ($p=0.023$)	survives both
LOXL2	+2.46 (0.014)	+1.41 (0.16)	+2.20 (0.028)	prolif-independent, dosage-driven
CDCA2, PBK, ESCO2	significant	—	≈ 0 (NS)	proliferation passengers

STC1 responds to its own copy number (deletion lowers its expression) and is up in progressed tumours beyond both dosage and proliferation. STC1 (stanniocalcin-1) is a secreted, hypoxia-induced factor promoting EMT, invasion and angiogenesis; LOXL2 is an ECM-crosslinking EMT driver — both biologically coherent with metastasis.

4. High STC1 predicts worse overall survival, independent of proliferation

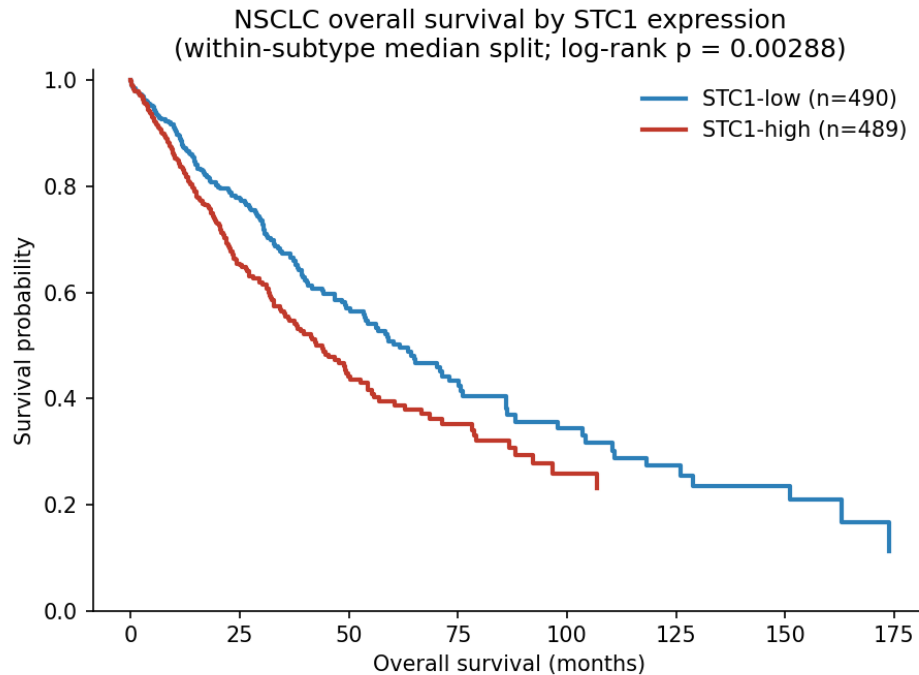


Figure 1. Kaplan–Meier overall survival by STC1 expression in NSCLC

Patients with high STC1 (within-subtype median split) had significantly shorter overall survival: **median 44.0 vs 61.6 months, log-rank p = 0.003**. The association held in a subtype-stratified Cox score test ($z = 2.56$, $p = 0.011$) and **survived proliferation adjustment** ($z = 2.27$, $p = 0.023$). Critically, the proliferation marker MKI67 showed the opposite behaviour — a nominal OS association ($p = 0.069$) that **vanished** under the same proliferation adjustment ($p = 0.55$) — confirming that STC1's prognostic value is not merely "proliferative tumours die faster."

STC1 thus rests on three concordant lines of evidence: (1) it lies on 8p, deleted in indolent but retained in metastatic tumours; (2) its expression is elevated in progressed tumours beyond dosage and proliferation; and (3) it is prognostic for worse survival, again beyond proliferation.

Discussion

Reframing metastasis as negative selection — asking what indolent tumours lose rather than what metastatic tumours gain — recovered a signal where conventional differential analysis found only proliferation. The result nominates **8p retention, and specifically STC1**, as a marker of metastatic capability in NSCLC, consistent with STC1's known pro-metastatic biology.

Limitations. (1) This is an **association, not causation**: STC1 is hypoxia-induced, so its elevation in aggressive tumours may partly reflect a hypoxic microenvironment rather than a cell-intrinsic driver. (2) The candidate genes sit on 8p21 while the deletion peak is 8p23.1; whole-arm loss removes both, so expression — not deletion — does the pointing. (3) TCGA samples are resected **primaries**; a definitive test requires **matched**

primary-metastasis tissue (e.g. MET500, MSK-MET), where the prediction is direct: STC1 expression, and 8p retention, should predict which primaries metastasize. (4) The distant-met deletion contrast rests on a sparse cell (1/124 deleted), mitigated by the solid indolent-group rate (~19/208).

Conclusion

In TCGA NSCLC, deep deletion of chromosome 8p marks indolent (true stage I) disease and is almost absent in tumours that metastasize. Expression narrows this arm-level signal to **STC1**, whose high expression independently predicts worse survival. We propose *8p/STC1 retention* as a testable marker of metastatic capability in NSCLC.

Reproducibility

```
python scripts/23_true_stage_i_lost_genes.py --save-dir results # 8p lost in true stage I
python scripts/24_chr8p23_which_gene.py --save-dir results # resolve arm -> STC1
python scripts/25_stc1_survival_check.py --save-dir results # STC1 vs OS
(proliferation-adjusted)
python scripts/26_stc1_km_curve.py --gene STC1 --save-dir docs/figures # KM figure
```

Full methods and intermediate results: nsclc_metastasis_spared_deletions.md,
nsclc_metastasis_expression.md.